Kwame Nkrumah University of
Science & Technology, Kumasi, Ghana

Introduction to Fluid Mechanics
ME 251

Fluid Statics

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Areas to Cover

- Fluid Statics
  - Hydrostatic forces on submerged surfaces
  - Buoyancy and flotation

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FLUID STATICS

- **Fluid statics** deals with problems associated with fluids at rest.

- Fluid statics is generally referred to as:

  ✓ **Hydrostatics** when the fluid is a liquid and

  ✓ **Aerostatics** when the fluid is a gas.

- No relative motion between adjacent fluid layers, and thus there are no shear (tangential) stresses in the fluid trying to deform it.

- The only stress to deal with is the normal stress -**pressure**.

  ✓ the variation of pressure is due only to the **weight** of the fluid.

HYDROSTATIC FORCES ON SUBMERGED PLANE SURFACES

- A plate exposed to a liquid is subjected to fluid pressure distributed over its surface.

- On a plane surface, the hydrostatic forces form a system of parallel forces.

  – The resultant force acts through the **Centre of Pressure**.

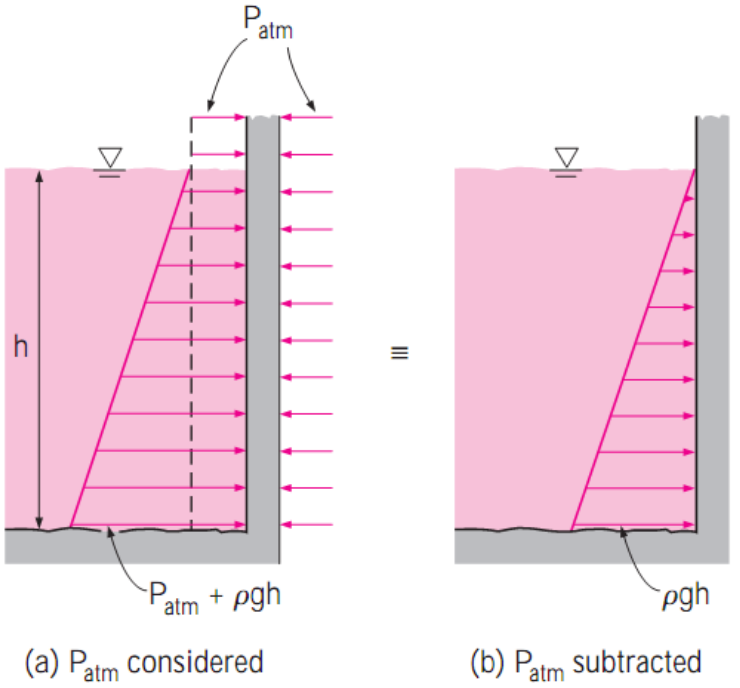




HYDROSTATIC FORCES ON SUBMERGED PLANE SURFACES

- In the Figure shown, the other side of the plate/wall may be open to the atmosphere (such as the dry side of a gate),
- Atmospheric pressure acts on both sides of the plate, yielding a zero resultant.
- It is convenient to work with the gauge pressure only .
- At the bottom of the lake:

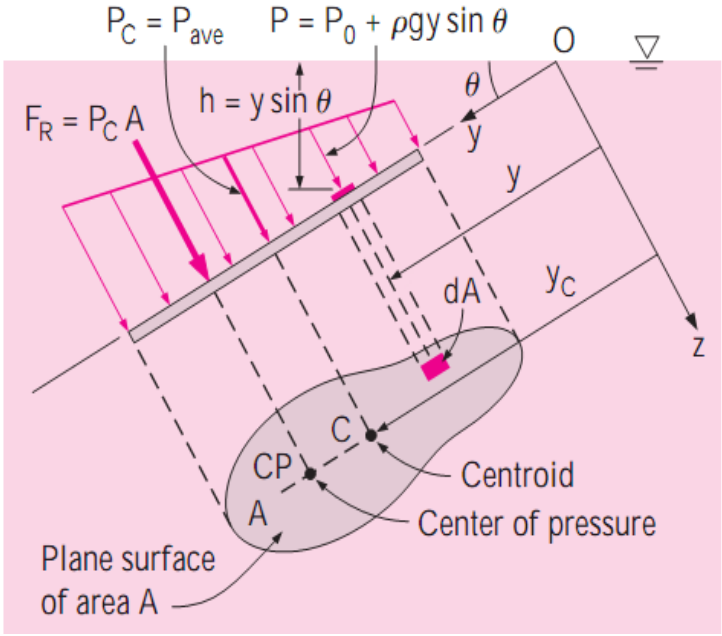
$$\mathbf{P_{gauge} = \rho gh}.$$



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HYDROSTATIC FORCES ON SUBMERGED PLANE SURFACES

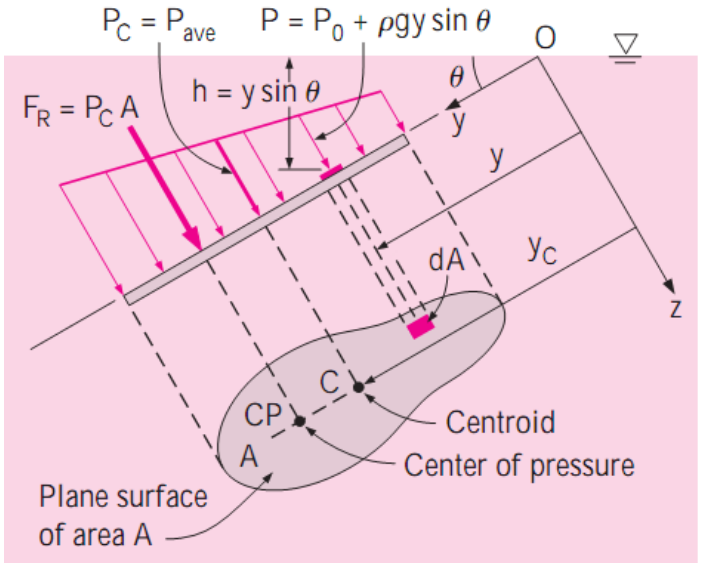
- A flat plate of arbitrary shape is completely submerged in a liquid.
- The plane of the surface intersects the horizontal free surface with an angle θ.
- The absolute pressure above the liquid is $\mathbf{P_0}$, which is the local atmospheric pressure $\mathbf{P_{atm}}$.
- Then the absolute pressure at any point on the plate is:



$$P = P_0 + \rho g h = P_0 + \rho g y \sin\theta$$

HYDROSTATIC FORCES ON SUBMERGED PLANE SURFACES

- The **resultant hydrostatic force F_R** acting on the surface is determined by integrating the force **PdA** acting on a differential area **dA** over the entire surface area,



$$F_R = \int_A P\, dA = \int_A (P_0 + \rho g y \sin\theta)\, dA = P_0 A + \rho g \sin\theta \int_A y\, dA$$

first moment of area $\int_A y\, dA$ $\qquad y_C = \frac{1}{A}\int_A y\, dA$

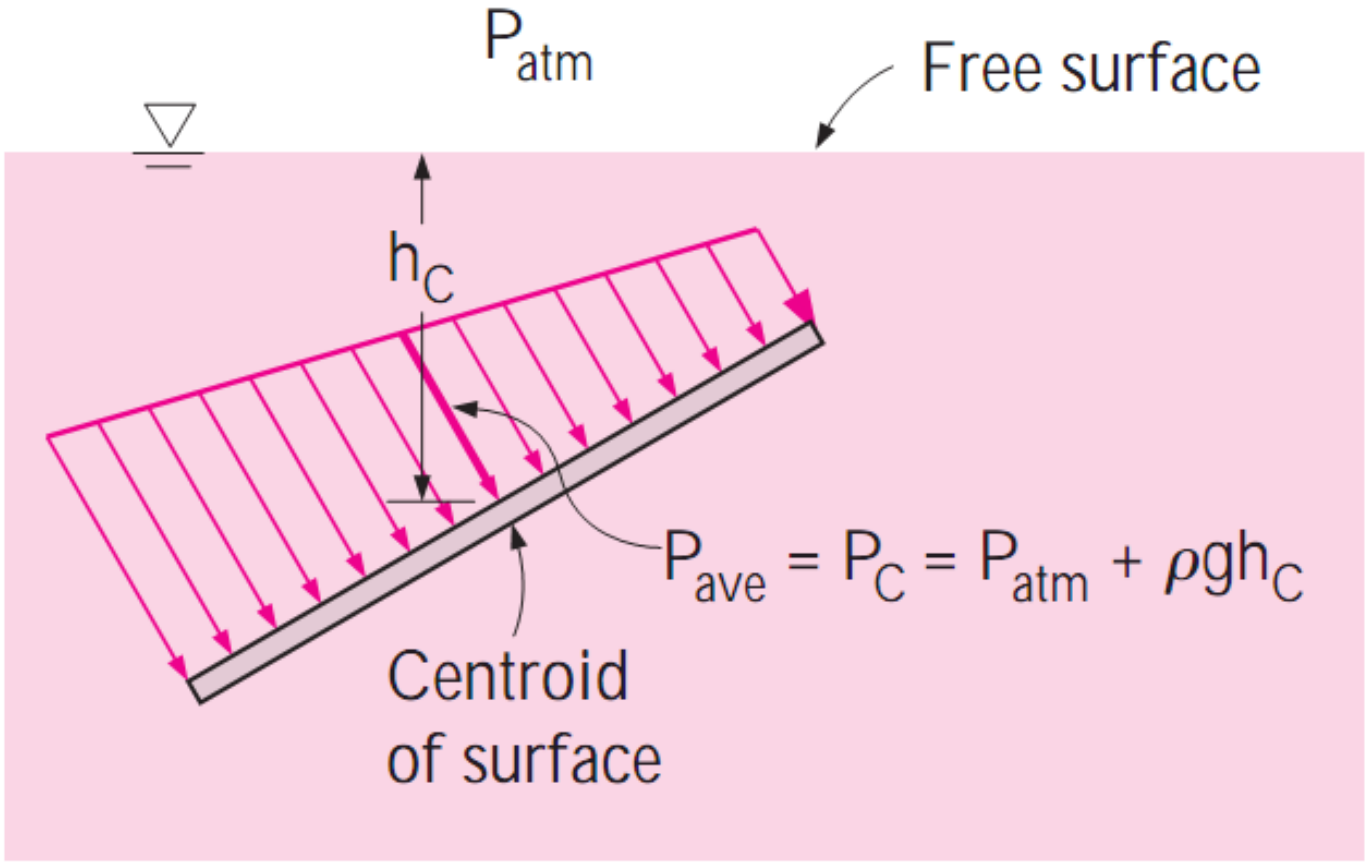
y_c is the y-coordinate of the centroid

$$F_R = (P_o + \rho g y_c \sin\theta)A \qquad = (P_o + \rho g h_c)A \qquad = P_{ave}A$$

- where $\mathbf{P_{ave}}$ or $\mathbf{P_C = P_0 + \varrho g h_C}$ is the pressure at the **centroid** of the surface, and is equivalent to the average pressure on the surface, and
- $\mathbf{h_C = y_C \sin\theta}$ is the vertical distance of the centroid from the free surface of the liquid.

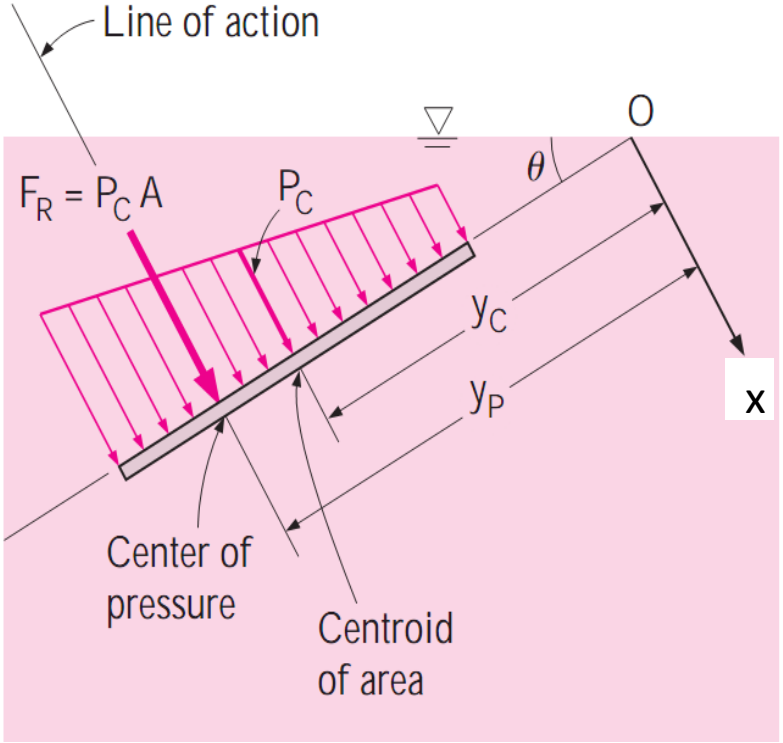
HYDROSTATIC FORCES ON SUBMERGED PLANE SURFACES

$$F_R = (P_o + \rho g y_c \sin\theta)A \qquad = (P_o + \rho g h_c)A \qquad = P_{ave}A$$



HYDROSTATIC FORCES ON SUBMERGED PLANE SURFACES

The **magnitude** of the **resultant force** acting on a plane surface of a completely submerged plate in a homogeneous (constant density) fluid is equal to **the product of the pressure P_C at the centroid of the surface and the area A of the surface**.



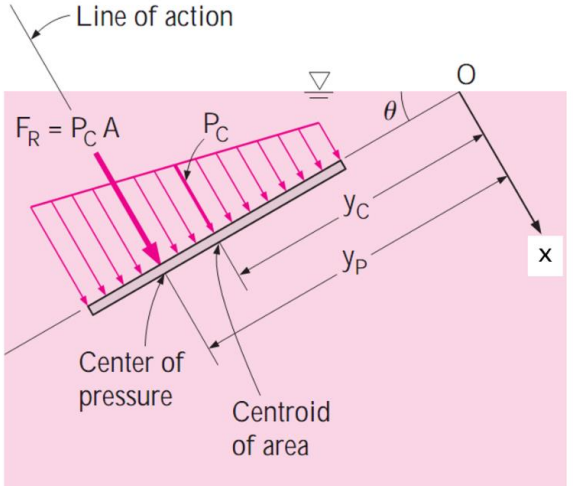
HYDROSTATIC FORCES ON SUBMERGED PLANE SURFACES

Two parallel force systems are equivalent if they have the **same magnitude** and the **same moment** about a given point.

The **line of action** of the **resultant hydrostatic force**, in general, does not pass through the centroid of the surface. It lies underneath where the pressure is higher.

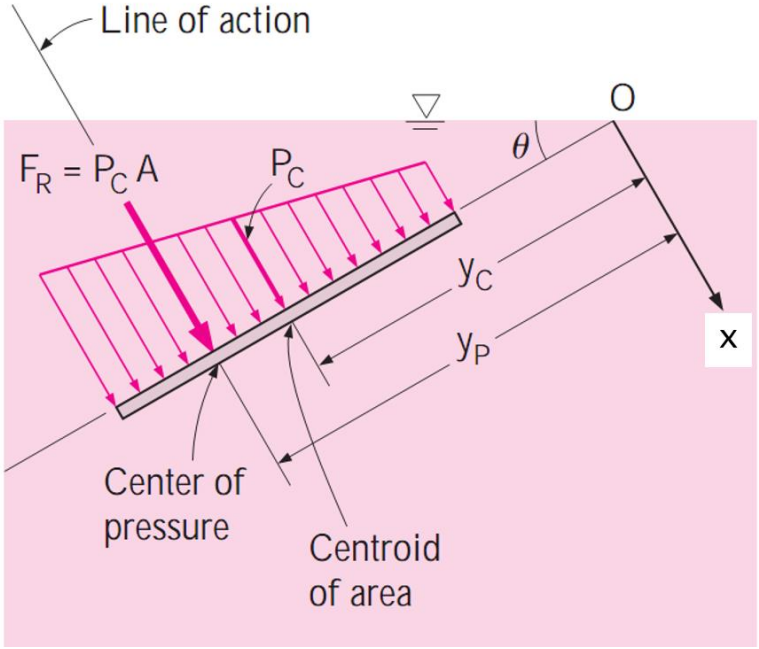
The point of intersection of the line of action of the resultant force and the surface is the **center of pressure**.

The vertical location of the line of action is determined by equating the **moment of the resultant force** to the **moment of the distributed pressure force** about the x-axis.



HYDROSTATIC FORCES ON SUBMERGED PLANE SURFACES

$$y_P F_R = \int_A yP \, dA$$



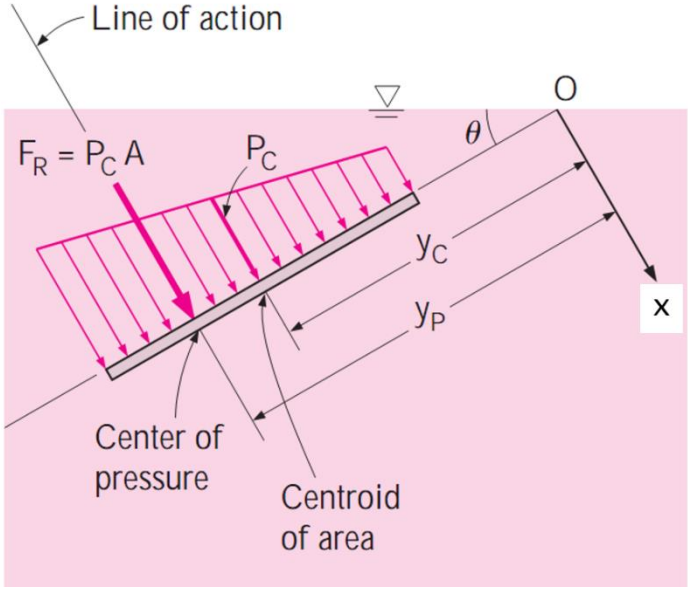
$$y_P F_R = P_0 y_C A + \rho g \sin\theta \, I_{xx,O}$$

$$I_{xx,O} = \int_A y^2 \, dA$$

Second moment of area (area moment of inertia) about the x-axis

HYDROSTATIC FORCES ON SUBMERGED PLANE SURFACES

- The **second moment** of area about **two parallel axes** are related to each other by the **parallel axis theorem**, which in this case is expressed as:



$$I_{xx,\,O} = I_{xx,\,C} + y_C^2 A$$

where $I_{xx,\,C}$ is the second moment of area about the x-axis passing **through the centroid** of the area and y_C (the y-coordinate of the centroid) is the distance between the two parallel axes.

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HYDROSTATIC FORCES ON SUBMERGED PLANE SURFACES

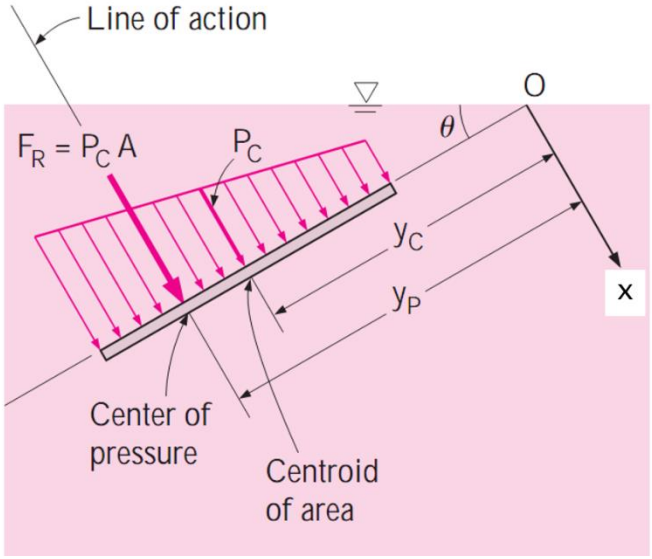
- Substituting the F_R relation and the $I_{xx,O}$ relation and solving for y_P

$$F_R = (P_o + \rho g y_c \sin\theta)A$$

$$I_{xx,O} = I_{xx,C} + y_C^2 A$$

$$y_p F_R = P_o y_c A + \rho g \sin\theta \, I_{XX,O}$$

$$y_P = y_C + \frac{I_{xx,C}}{[y_C + P_0/(\rho g \sin\theta)]A}$$

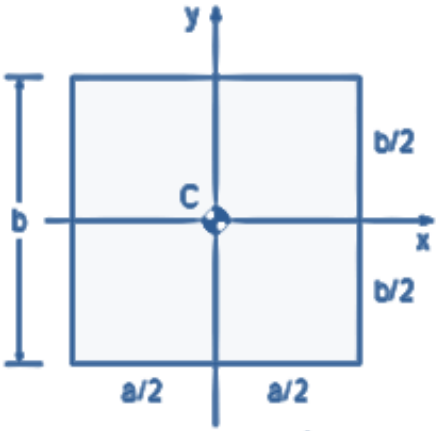


When P_o is zero:

$$y_P = y_C + \frac{I_{xx,C}}{y_C A}$$

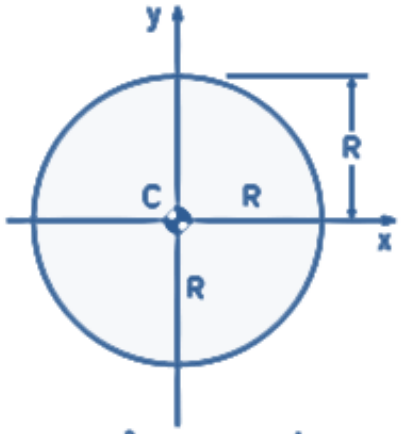
Knowing y_P, the vertical distance of the center of pressure from the free surface is determined from $h_P = y_P \sin\theta$.

The Centroid and the Centroidal Moments of Inertia for Some Common Geometries



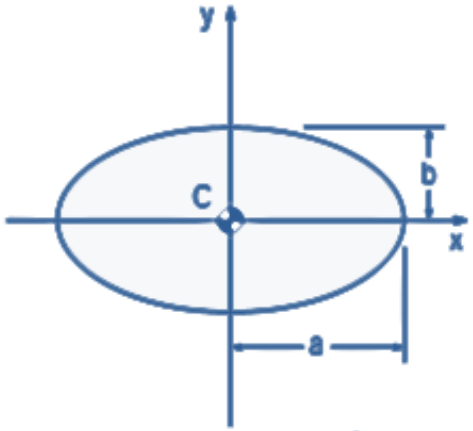
$A = ab,\ I_{xx,C} = ab^3/12$

(a) Rectangle



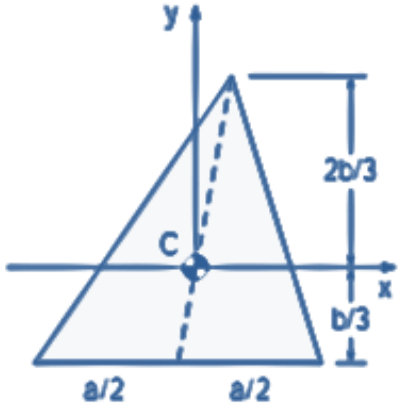
$A = \pi R^2,\ I_{xx,C} = \pi R^4/4$

(b) Circle



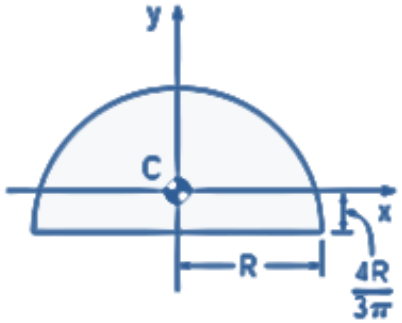
$A = \pi ab,\ I_{xx,C} = \pi ab^3/4$

(c) Ellipse



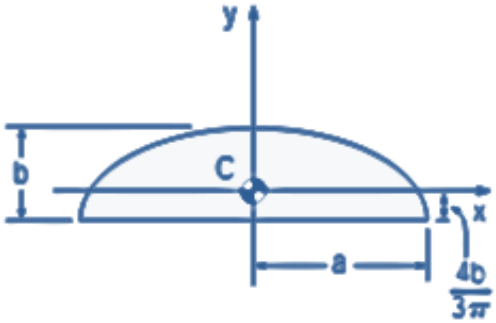
$A = ab/2,\ I_{xx,C} = ab^3/36$

(d) Triangle



$A = \pi R^2/2,\ I_{xx,C} = 0.109757R^4$

(e) Semicircle



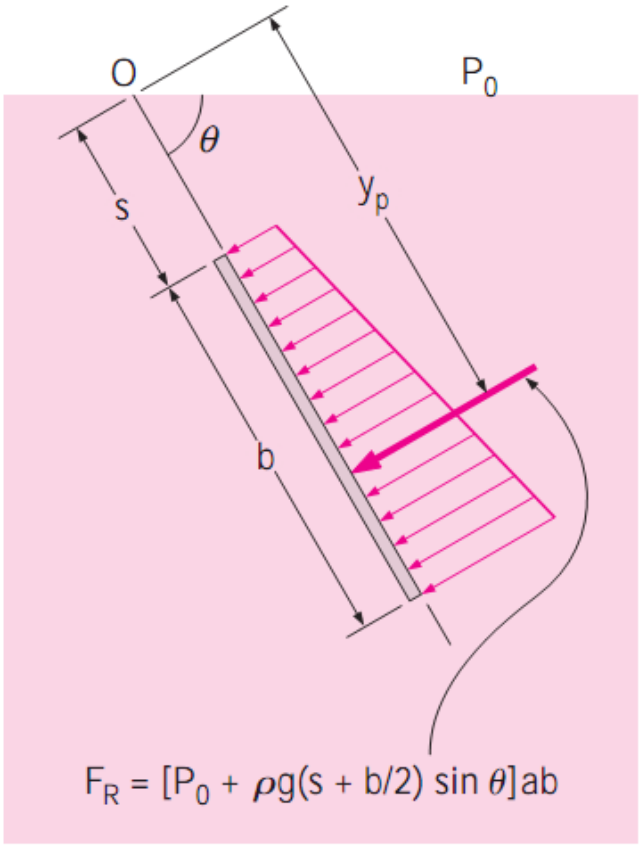
$A = \pi ab/2,\ I_{xx,C} = 0.109757ab^3$

(f) Semiellipse

Special Case: Submerged Rectangular Plate

- Consider a completely submerged rectangular flat plate of **height b** and **width a** tilted at an **angle θ** from the horizontal and whose top edge is horizontal and is at a **distance s** from the free surface along the plane of the plate.

- The resultant **hydrostatic force** on the upper surface is equal to the **average pressure**, which is the pressure at the midpoint of the surface, times the **surface area A**. That is:



(a) Tilted plate

$$F_R = P_C A = [P_0 + \rho g(s + b/2) \sin\theta]ab$$

Submerged Rectangular Plate

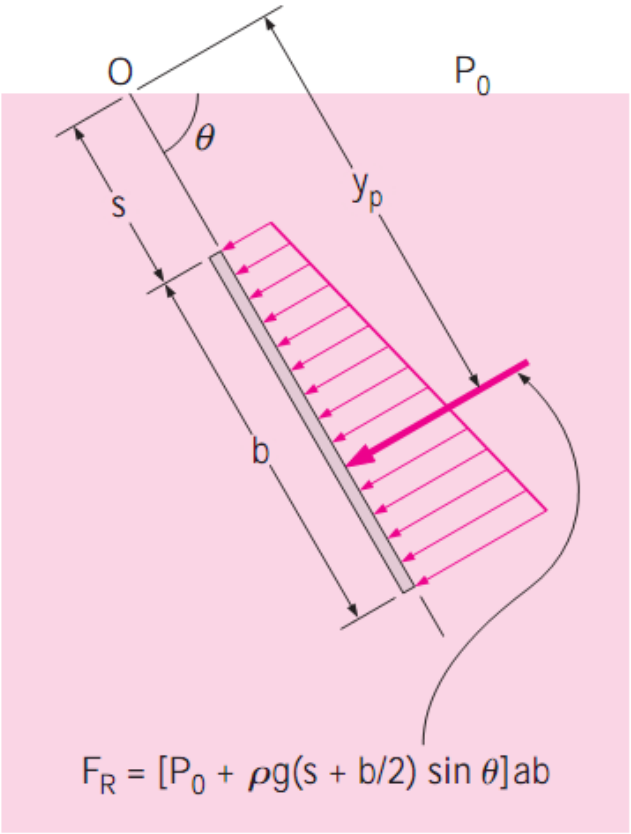
- $\mathbf{F_R}$ acts at a vertical distance of $\mathbf{h_P = y_P \sin\theta}$ from the free surface directly beneath the centroid of the plate where,

$$y_P = s + \frac{b}{2} + \frac{ab^3/12}{[s + b/2 + P_0/(\rho g \sin\theta)]ab}$$

$$= s + \frac{b}{2} + \frac{b^2}{12[s + b/2 + P_0/(\rho g \sin\theta)]}$$

When the upper edge of the plate is at the free surface and s = 0, then:

$$F_R = [P_0 + \rho g(b \sin\theta)/2]ab$$



(a) Tilted plate

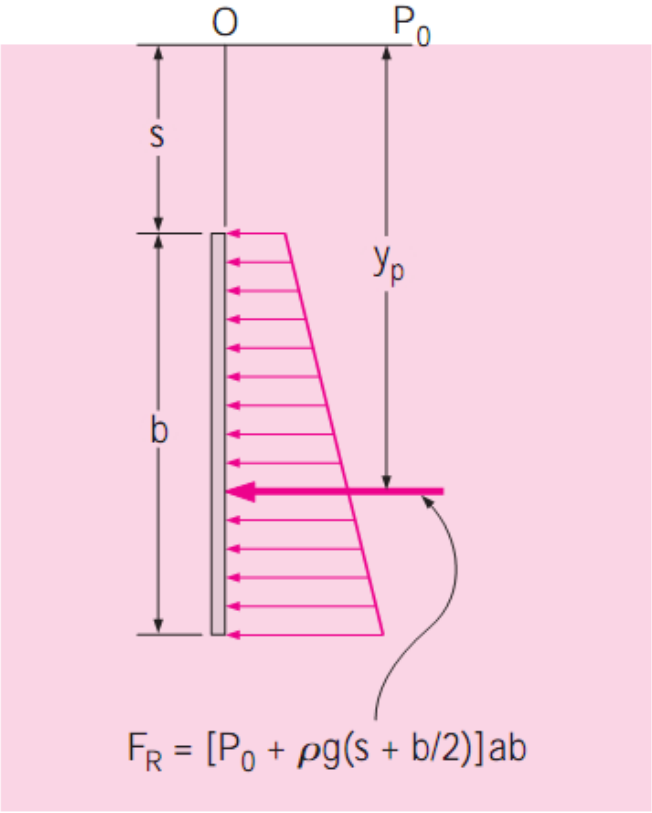
Submerged Rectangular Plate

- For a completely submerged vertical plate ($\theta = 90°$) whose top edge is horizontal, the hydrostatic force can be obtained by setting $\sin\theta = 1$

$$F_R = [P_0 + \rho g(s + b/2)]ab$$

When S = 0

$$F_R = (P_0 + \rho g b/2)ab$$



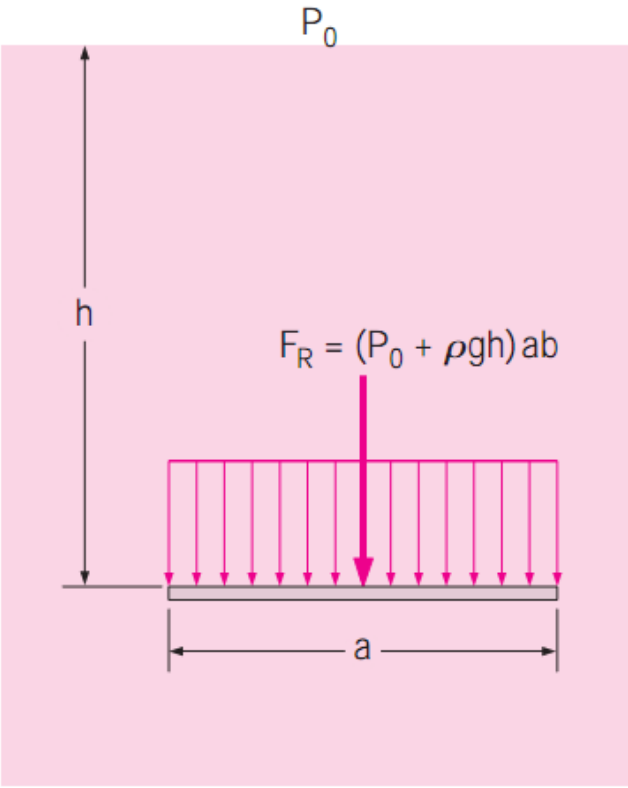
(b) Vertical plate

Submerged Rectangular Plate

- The pressure distribution on a submerged horizontal surface is uniform,
- and its magnitude is $\mathbf{P = P_0 + \rho g h}$, where **h** is the distance of the surface from the free surface.
- Therefore, the hydrostatic force acting on a horizontal rectangular surface is:

$$F_R = (P_0 + \rho g h)ab$$

- and it acts through the midpoint of the plate.



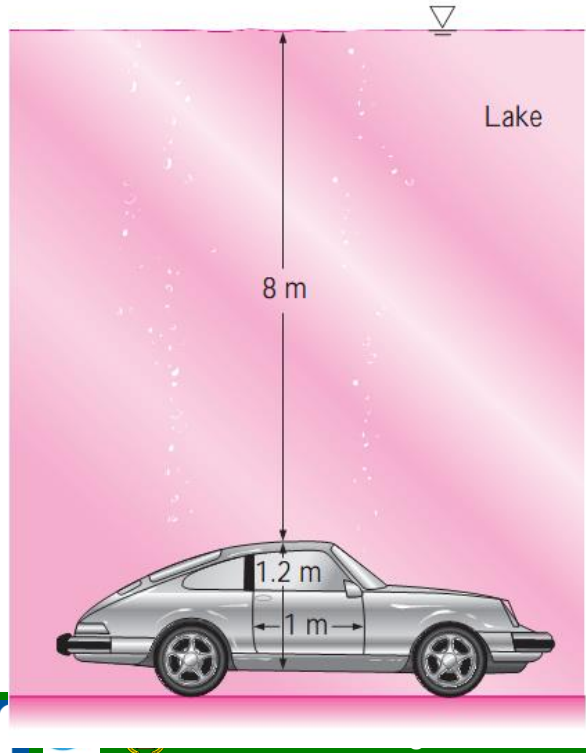
(c) Horizontal plate

Question

A heavy car plunges into a lake during an accident and lands at the bottom of the lake on its wheels. The door is 1.2 m high and 1 m wide, and the top edge of the door is 8 m below the free surface of the water. Determine the hydrostatic force on the door and the location of the pressure center, and discuss if the driver can open the door.

Assumptions:

1. The bottom of the lake is horizontal.
2. The passenger cabin is well-sealed and there is no water leakage inside.
3. The door can be approximated as a vertical rectangular plate.
4. The pressure in the car remains at atmospheric pressure. Atmospheric pressure cancells out, since it acts on both sides of the door.



Question

The average pressure on the door is the pressure at the centroid (midpoint) of the door:

$$P_{ave} = P_c = \rho g h_c = \rho g\left(s + \frac{b}{2}\right)$$

$$= \left(\frac{1000\ \text{kg}}{\text{m}^3}\right)\left(\frac{9.8\ \text{m}}{\text{s}^2}\right)\left(8\ \text{m} + \frac{1.2\ \text{m}}{2}\right) = 84.4\ \text{kN/m}^2$$

The resultant hydrostatic force (F_R) is:

$$F_R = P_{ave}A = (84.4\ \text{kN/m}^2)(1\ \text{m x}\ 1.2\ \text{m}) = 101.3\ \text{kN}$$

The line of action of the resultant force, y_p is found as:

$$= s + \frac{b}{2} + \frac{b^2}{12[s + b/2 + P_0/(\rho g \sin\theta)]}$$

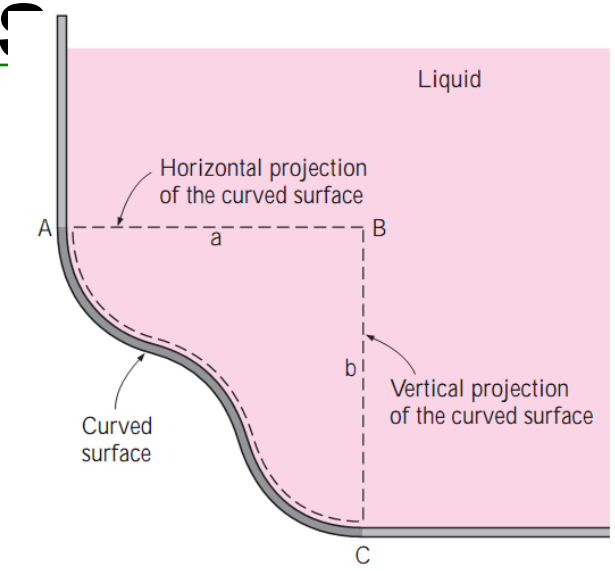
Setting $P_o = 0$

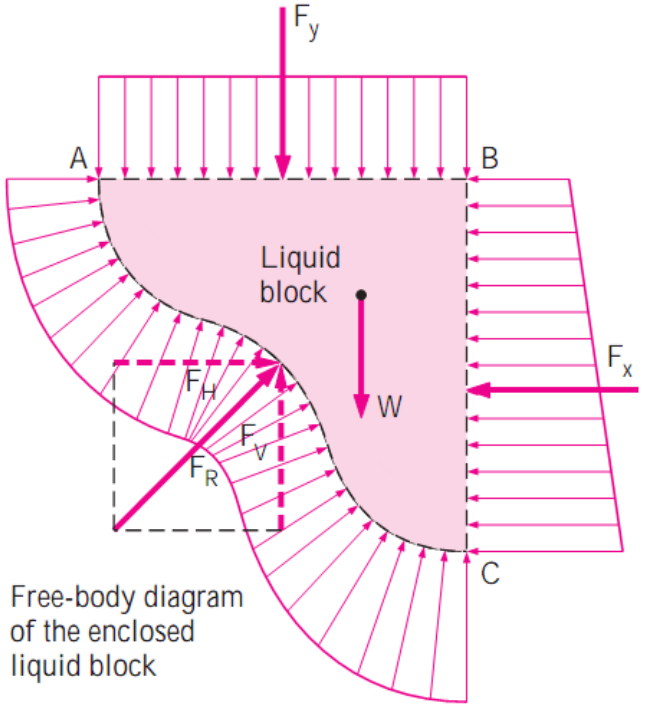
$$y_p = s + \frac{b}{2} + \frac{b^2}{12(s + b/2)} = 8 + \frac{1.2}{2} + \frac{1.2^2}{12(8 + 1.2/2)} = 8.61\ \text{m}$$

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HYDROSTATIC FORCES ON SUBMERGED CURVED SURFACES

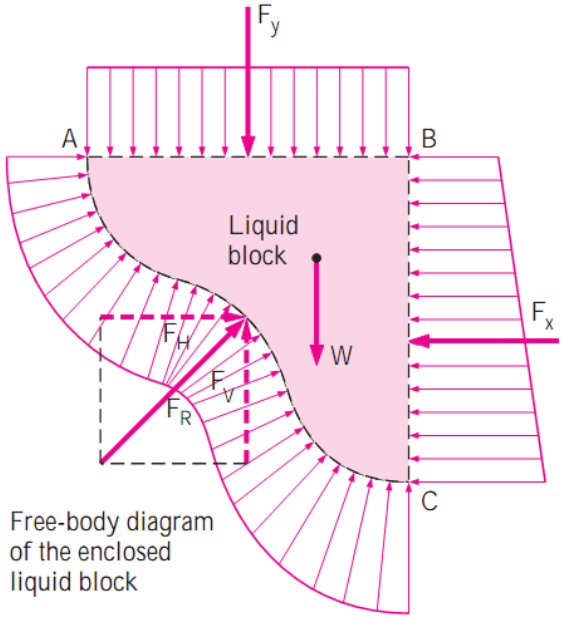
- The resultant hydrostatic force F_R comprises the horizontal F_H and vertical F_V components.
-
- The vertical surface of the liquid block considered is simply the projection of the curved surface on a vertical plane.
- The horizontal surface is the projection of the curved surface on a horizontal plane.
- The **resultant force** acting on the curved solid surface is then **equal and opposite** to the force acting on the curved liquid surface (Newton's third law).





SUBMERGED CURVED SURFACES

- The weight of the enclosed liquid block of volume V is simply **W = ρgV**.
- It acts downward through the centroid of this volume.
- The fluid block is in static equilibrium, the force balances in the horizontal and vertical directions give.



Horizontal force component on curved surface:	$F_H = F_x$
Vertical force component on curved surface:	$F_V = F_y + W$

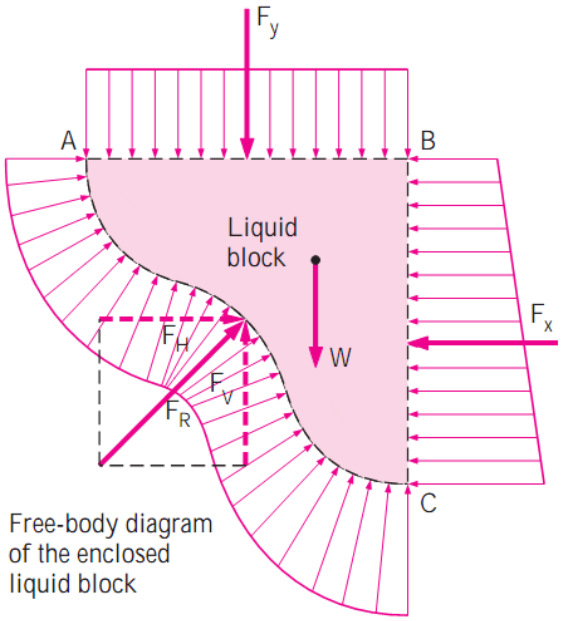
Where the summation **F_y + W** is a vector addition (i.e., add magnitudes if both act in the same direction and subtract if they act in opposite directions).

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SUBMERGED CURVED SURFACES

The **horizontal component** of the hydrostatic force acting on a curved surface is equal (in both magnitude and the line of action) to the hydrostatic force acting on the **vertical projection** of the curved surface.

The **vertical component** of the hydrostatic force acting on a curved surface is equal to the hydrostatic force acting on the **horizontal projection** of the curved surface, plus (minus, if acting in the opposite direction) the weight of the fluid block.



SUBMERGED CURVED SURFACES

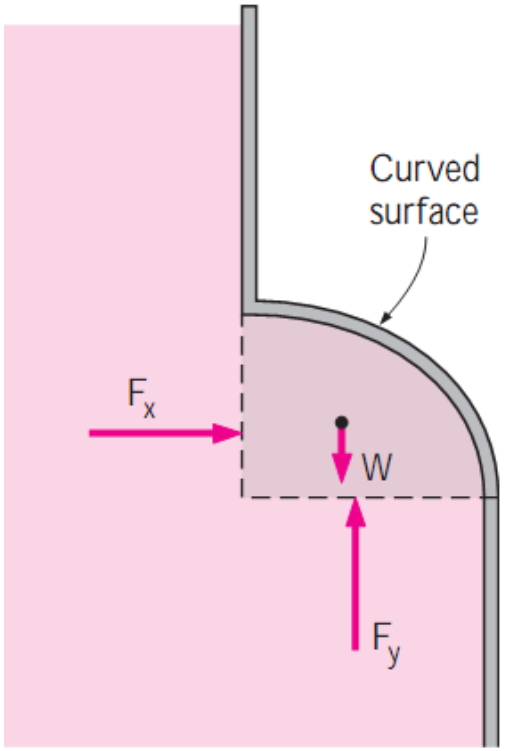
- The magnitude of the resultant hydrostatic force acting on the curved surface is:

$$F_R = \sqrt{F_H^2 + F_V^2}$$

- The angle (*a*) it makes with the horizontal is:

$$\tan \alpha = F_V / F_H$$

- NB: a curved surface above a liquid; the weight of the liquid is subtracted from the vertical component

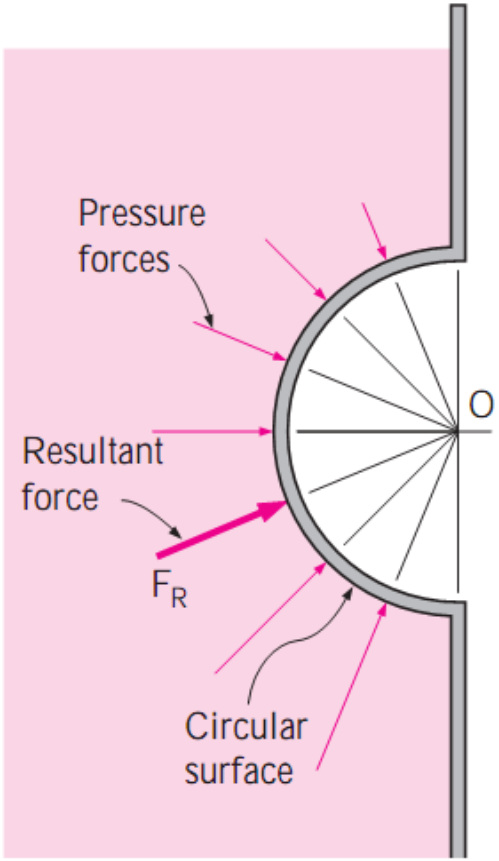


SUBMERGED CURVED SURFACES

- When the curved surface is a circular arc (full circle or any part of it), The resultant hydrostatic force acting on the surface always passes through the **center of the circle**.

  ✓ This is because the pressure forces are normal to the surface, and all lines normal to the surface of a circle pass through the center of the circle.

- Thus, the pressure forces form a concurrent force system at the center, which can be reduced to a single equivalent force at that point.



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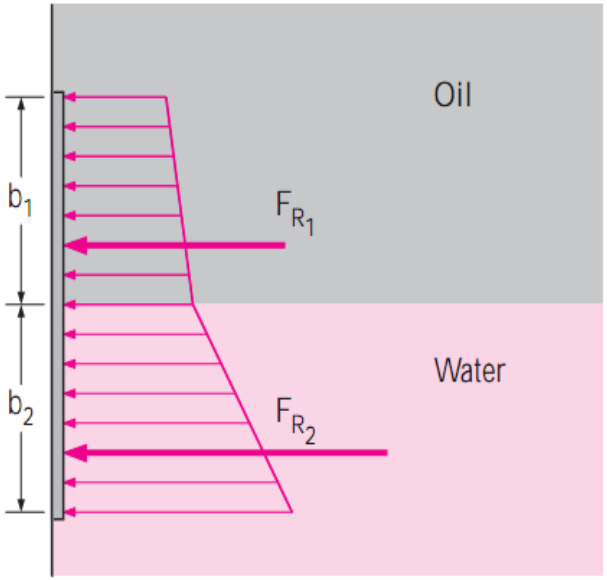
SUBMERGED CURVED SURFACES

- Hydrostatic forces acting on a plane or curved surface submerged in a **multilayered fluid** of different densities can be determined by

  ✓ considering different parts of surfaces in different fluids as different surfaces,

  ✓ finding the force on each part, and

  ✓ Then adding them using vector addition.



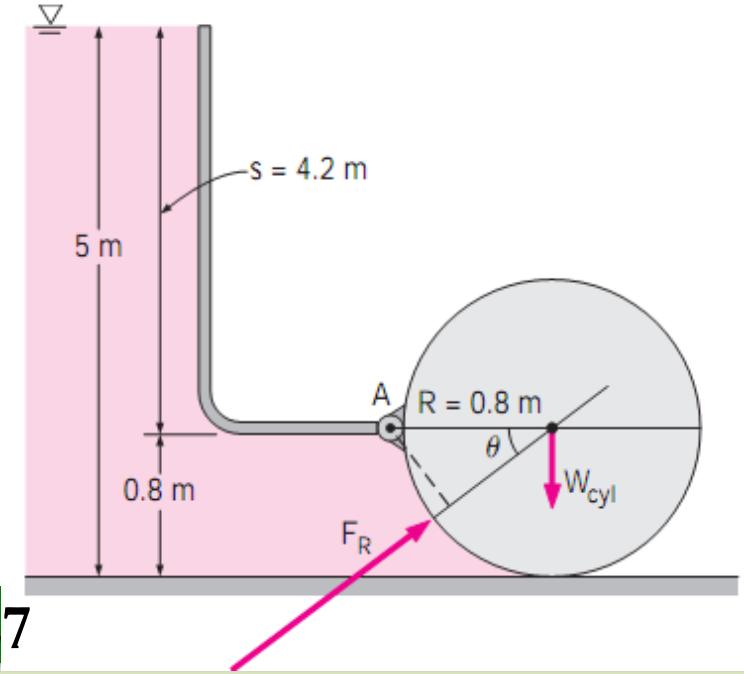
- For a plane surface, it can be expressed as

$$F_R = \sum F_{R,i} = \sum P_{C,i} A_i \qquad P_{C,i} = P_0 + \rho_i g h_{C,i}$$

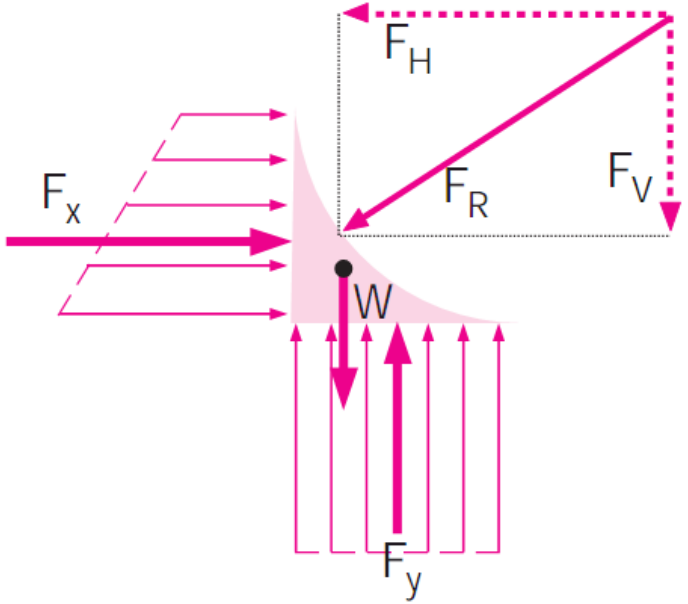
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QUESTION

A long solid cylinder of **radius 0.8 m** hinged at point A is used as an automatic gate. When the water level reaches 5 m, the gate opens by turning about the hinge at point A. Determine (a) **the hydrostatic force** acting on the cylinder and its **line of action** when the gate opens and (b) the weight of the cylinder per m length of the cylinder.



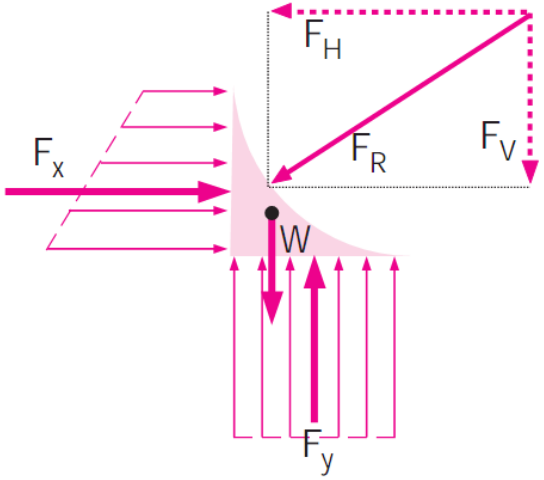
FBD of Schematic



QUESTION

Assumptions:

- Friction at the hinge is negligible.
- Atmospheric pressure acts on both sides and so cancels out.



$$F_H = F_X = P_{ave}A = \rho g h_c A$$
$$= \rho g(s + R/2)A$$

$$=(1000\ \text{kg/m}^3)(9.81\ \text{m/s}^2)(4.2\ \text{m}+0.8/2\ \text{m})(0.8\ \text{m x}\ 1\text{m})$$

$$= 36.1\ \text{kN}$$

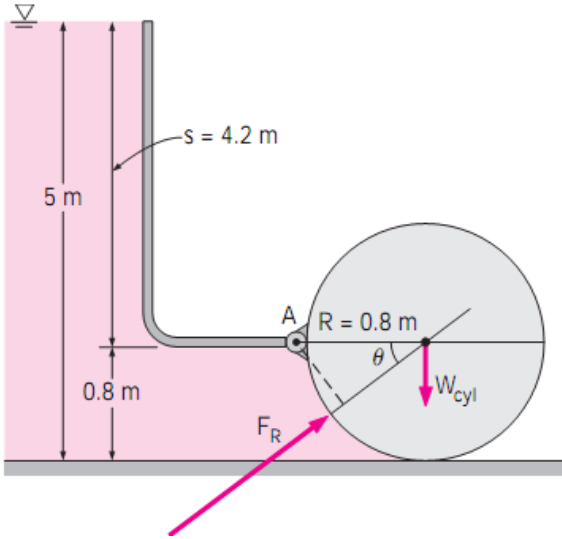
Vertical force on horizontal surface (upward)

$$F_y = P_{ave}A = \rho g h_c A = \rho g h_{bottom} A$$

$$=(1000\ \text{kg/m}^3)(9.81\ \text{m/s}^2)(5\ \text{m})(0.8\ \text{m x}\ 1\text{m})$$

$$= 39.2\ \text{kN}$$

QUESTION



Weight of fluid block per unit length (downward)

$$W = mg = \rho gV = \rho g\,(R^2 - \pi R^2/4)(1\text{ m})$$

$$= (1000\text{ kg/m}^3)(9.81\text{ m/s}^2)(0.8\text{ m})^2(1-\pi/4)(1\text{ m})$$

$$= 1.3\text{ kN}$$

The net vertical upward vertical force is:

$$F_v = F_y - W = 39.2 - 1.3 = 37.9\text{ kN}$$

The magnitude and direction of the hydrostatic force is:

$$F_R = \sqrt{F_H^2 + F_V^2} = \sqrt{36.1^2 + 37.9^2} = 52.3\text{ kN}$$

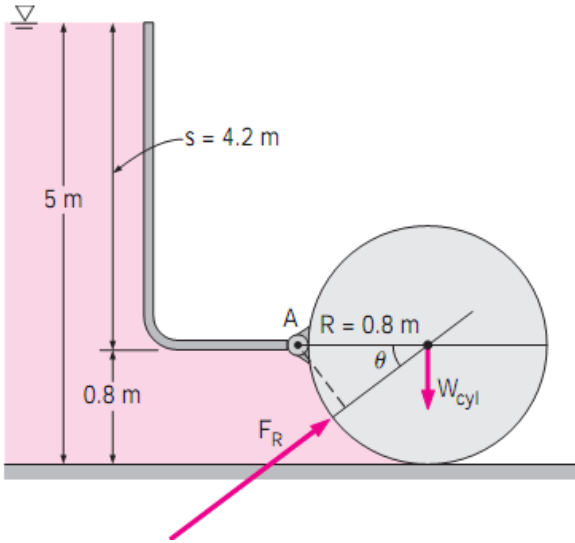
$$\tan\theta = {}^{F_V}/_{F_H} = \frac{37.9}{36.1} = 1.05$$

$$\theta = 46.4^\circ$$

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QUESTION

(b) When the water level is 5 m high, the gate is about to open and this the reaction force at the bottom of the cylinder is zero.



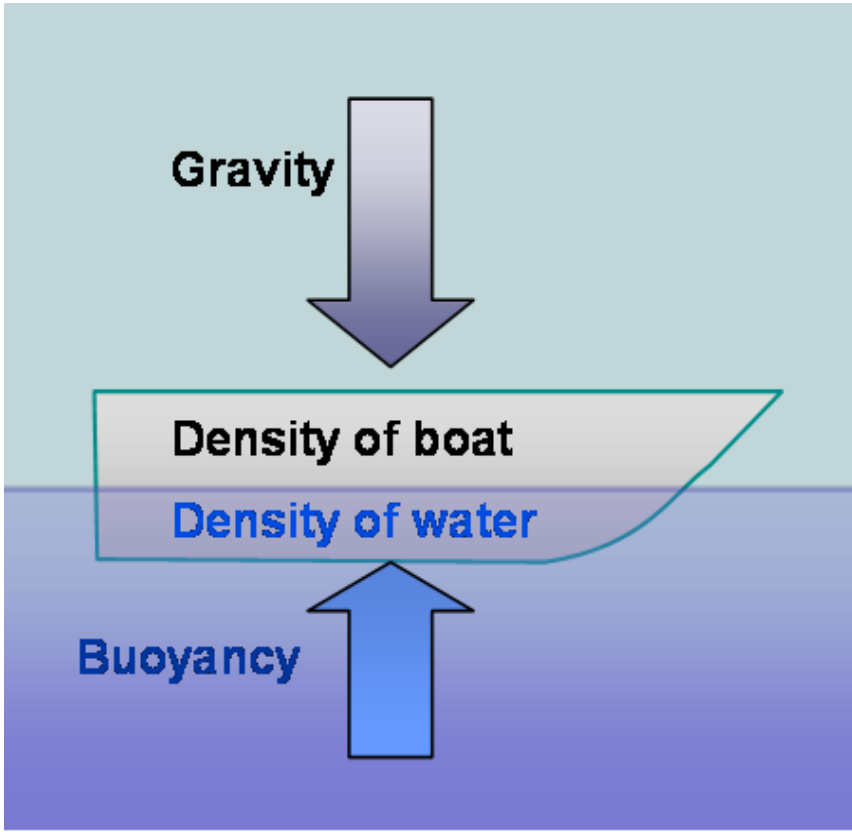
Taking moment about point A at the location of the hinge and equating it zero yields:

$$F_R R \sin\theta - W_{cycl} R = 0$$

$$W_{cycl} = F_R \sin\theta = (52.3\ \text{kN}) \sin 46.4^{o} = 37.9\ \text{kN}$$

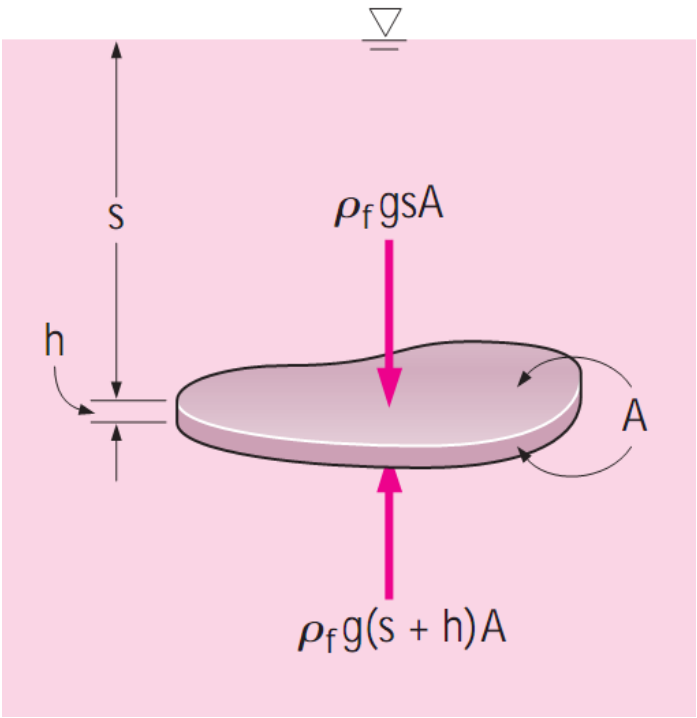
BUOYANCY AND STABILITY

- An object feels lighter and weighs less in a liquid than it does in air.
- An objects made of wood or other light materials float on water.
- A fluid exerts an upward force on a body immersed in it called the **buoyant** force.
- The buoyant force is caused by the increase of pressure in a fluid with depth.



BUOYANCY AND STABILITY

- The pressures at the top and bottom surfaces of the plate are $\rho_f gs$ and $\rho_f g(s+h)$, respectively.
- Force $F_{top}=\rho_f gsA$ acts downward on the top surface.
- Force $F_{bottom}=\rho_f g(s+h)A$ acts upward on the bottom surface.
-
- The difference between these two forces is a net upward force (the buoyant force).



$$F_B = F_{bottom} - F_{top} = \rho_f g(s + h)A - \rho_f gsA = \rho_f ghA = \rho_f gV$$

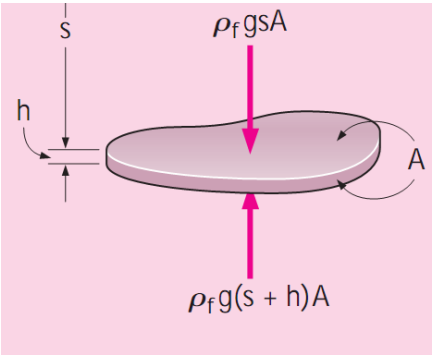
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BUOYANCY AND STABILITY

$$F_B = \rho_f g V$$

- **V = hA** is the volume of the plate.
- But the relation **$\rho_f gV$** is the weight of the liquid whose volume is equal to the volume of the plate.

  Buoyant force = weight of the liquid displaced

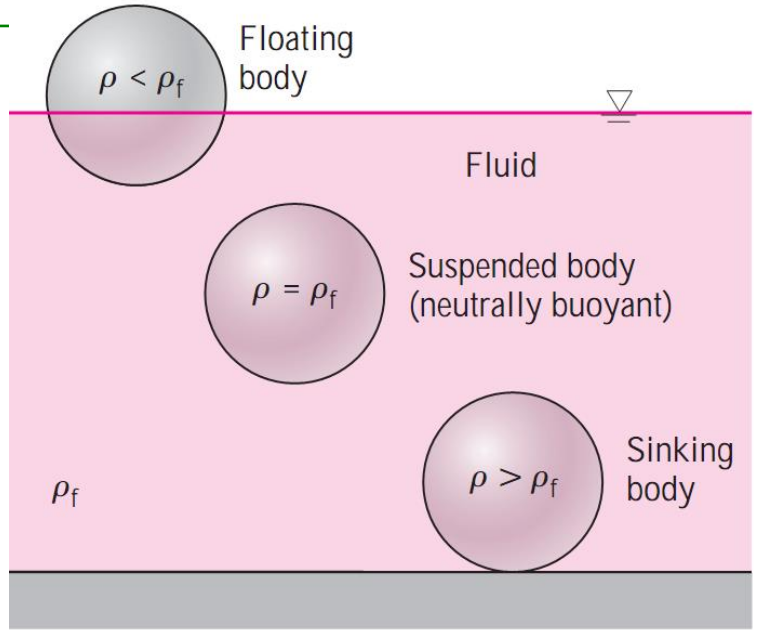


- The buoyant force is independent of:
  - ✓ the distance of the body from the free surface.
  - ✓ the density of the solid body.
- The **buoyant force** acting on a body **immersed** in a fluid is equal to the **weight of the fluid displaced by the body**, and it acts upward through the centroid of the displaced volume.

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BUOYANCY AND STABILITY

- For floating bodies:
  - the weight of the entire body must be equal to the buoyant force, (weight of the fluid whose volume is the volume of the submerged portion of the floating body).



Submerged volume fraction

$$F_B = W \rightarrow \rho_f g V_{sub} = \rho_{ave,\,body} g V_{total} \Rightarrow \frac{V_{sub}}{V_{total}} = \frac{\rho_{ave,\,body}}{\rho_f}$$

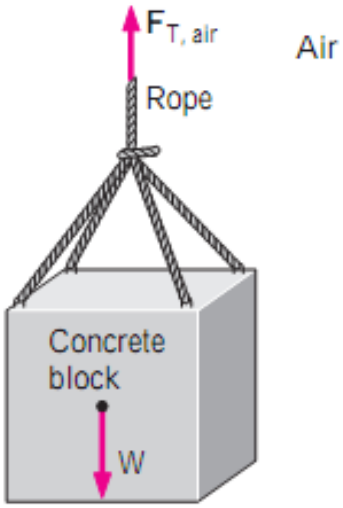
- ✓ The submerged volume fraction of a floating body is equal to the ratio of the average density of the body to the density of the fluid.
- ✓ When the density ratio is equal to or greater than one, the floating body becomes completely submerged.

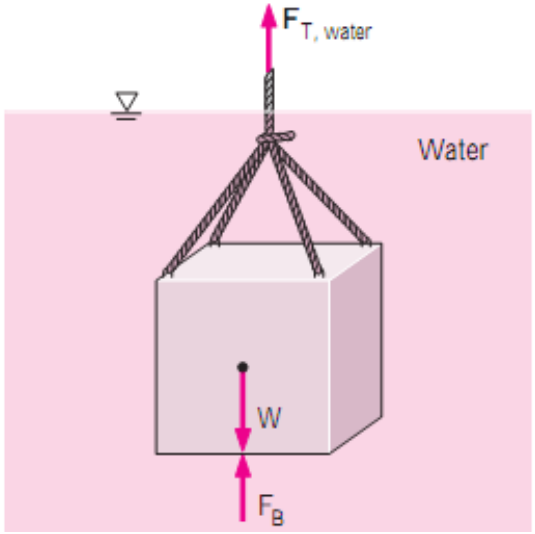
QUESTION

A crane is used to lower weights into the sea (density = 1025 kg/m^3) for an underwater construction project. Determine the tension in the rope of the crane due to a rectangular 0.4-m x 0.4-m x 3-m concrete block (density = 2300 kg/m^3) when it is

(a) suspended in the air and

(b) completely immersed in water.





Assumptions:

- Buoyancy of air is negligible
- The weight of the ropes is negligible

QUESTION

Volume of the block,

$V = (0.4\ m)\ (0.4\ m)(3\ m) = 0.48\ m^3$

Tension in rope (in air):

$F_{T,\ air}$ = Weight of concrete = $\rho_{concrete} gV$

$=(2300\ kg/m^3)(9.81\ m/s^2)(0.48\ m^3) = 10.8\ kN$

(b) When the block is immersed in water, buoyant force F_B acts on it (upward)

$F_B = \rho_f gV = (1025\ kg/m^3)(9.81\ m/s^2)(0.48\ m^3) = 4.8\ kN$

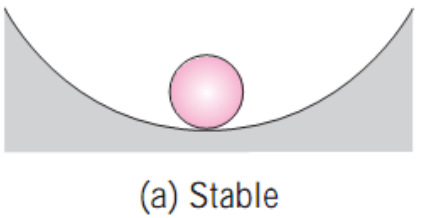
$F_{T,\ water} = W - F_B = 10.8 - 4.8 = 6.0\ kN$

Note: The weight of the concrete block decreases and thus the tension in the rope, decreases by (10.8 – 6.0)/10.8 = 44.4% in water.

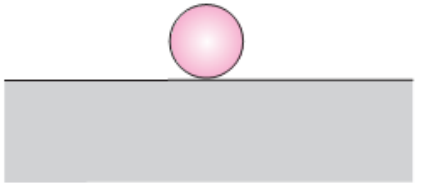
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Stability of Immersed and Floating Bodies

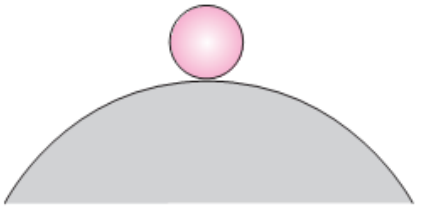
- An important application of the buoyancy concept is the assessment of the stability of immersed and floating bodies with no external attachments.
- Case (a) is **stable** since any small disturbance generates a restoring force (due to gravity) that returns it to its initial position.
- Case (b) is **neutrally stable** because if someone moves the ball to the right or left, it would stay put at its new location. It has no tendency to move back to its original location, nor does it continue to move away.
- Case (c) is a situation in which the ball may be at rest at the moment, but any disturbance, even an infinitesimal one, causes the ball to roll off the hill—it does not return to its original position; rather it diverges from it (**unstable**).



(a) Stable



(b) Neutrally stable

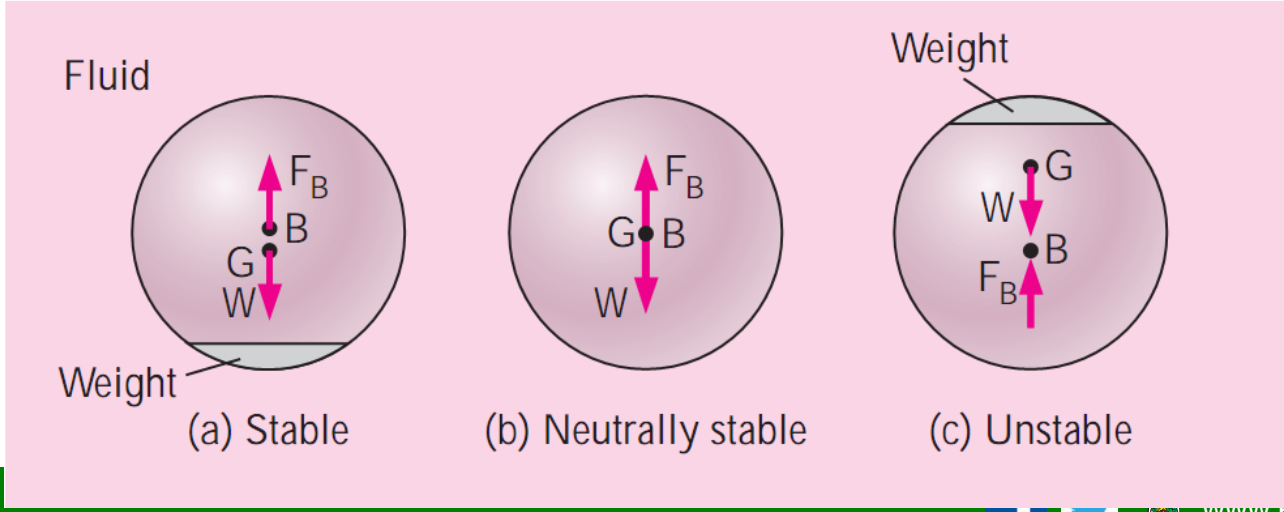


(c) Unstable

Rotational Stability

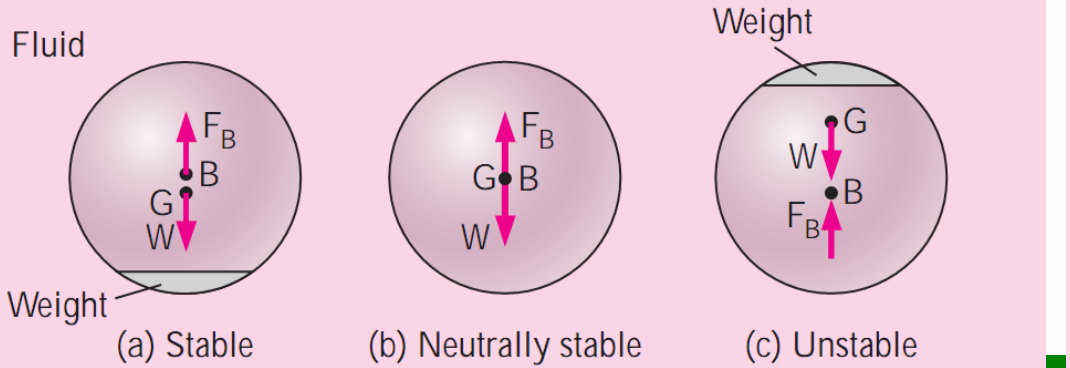
- The rotational stability of an immersed body depends on the relative locations of the **center of gravity, G** of the body and the **center of buoyancy, B** (which is the centroid of the displaced volume).

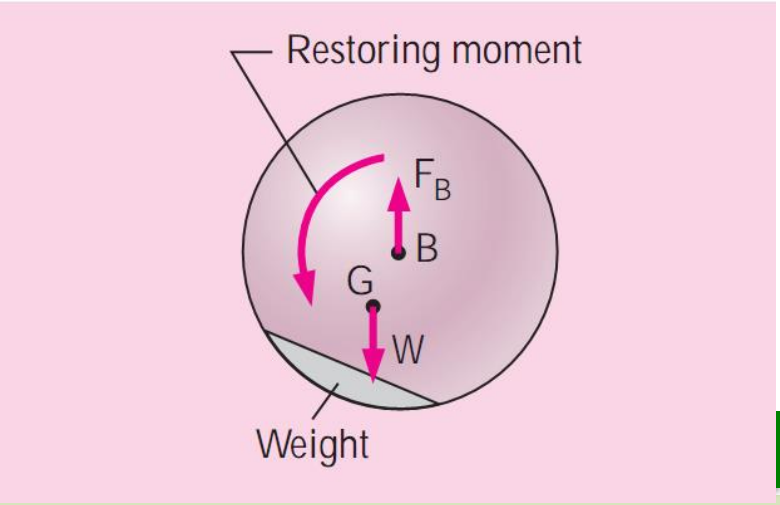
- An immersed body is stable if the body is bottom-heavy and thus **point G is directly below point B**.



Rotational Stability

- A rotational disturbance of a stable body produces a restoring moment to return the body to its original stable position.
  - A stable design for a submarine calls for the engines and the cabins for the crew to be located at the lower half in order to **shift the weight to the bottom** as much as possible.
  - Hot-air or helium balloons (which can be viewed as being immersed in air) are also stable since the cage that carries the load is at the bottom.

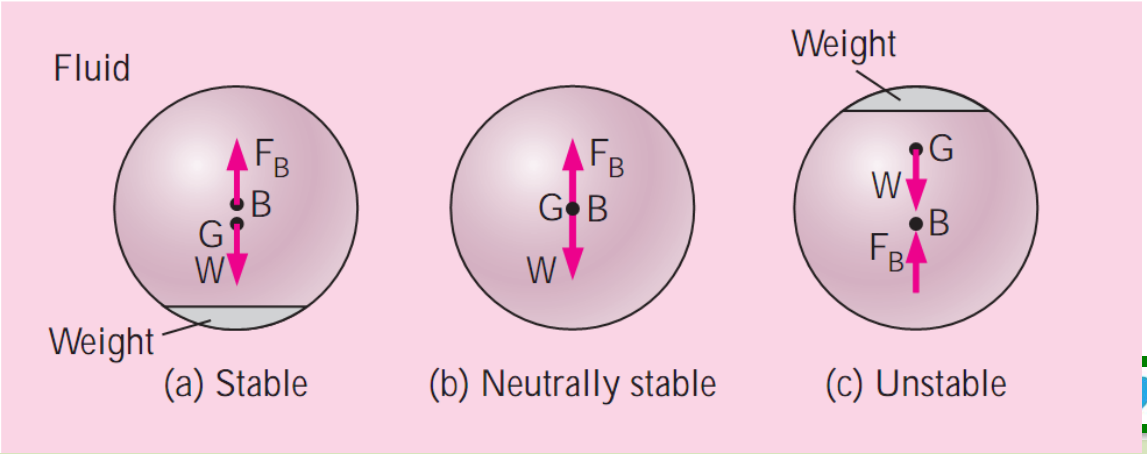




Rotational Stability

- An **immersed** body whose center of gravity G is directly **above point B is unstable**, and any disturbance will cause this body to turn upside down.

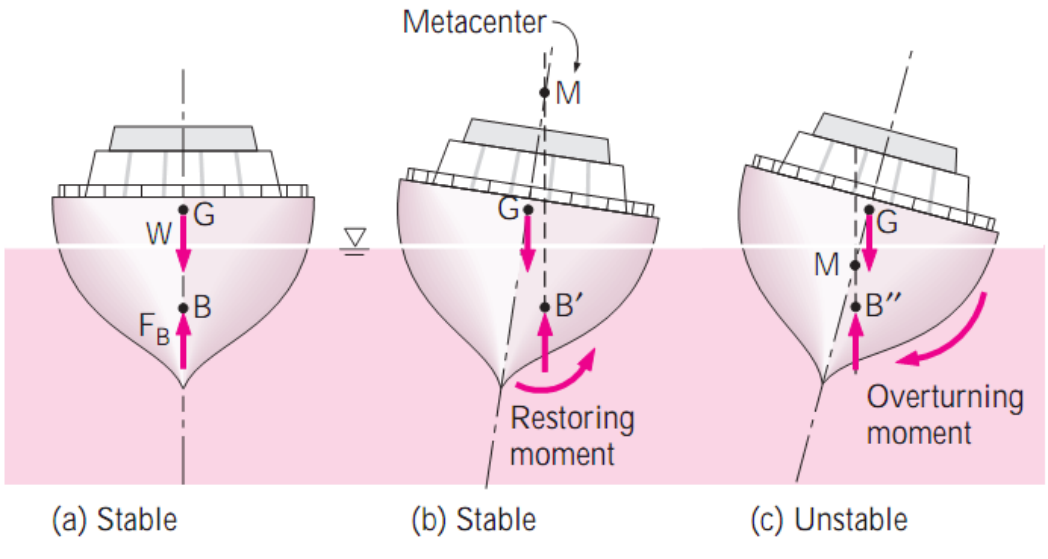
- A body for which G and B coincide is neutrally stable. For neutrally stable bodies, there is no tendency to overturn or right themselves.



Stability of Floating Bodies

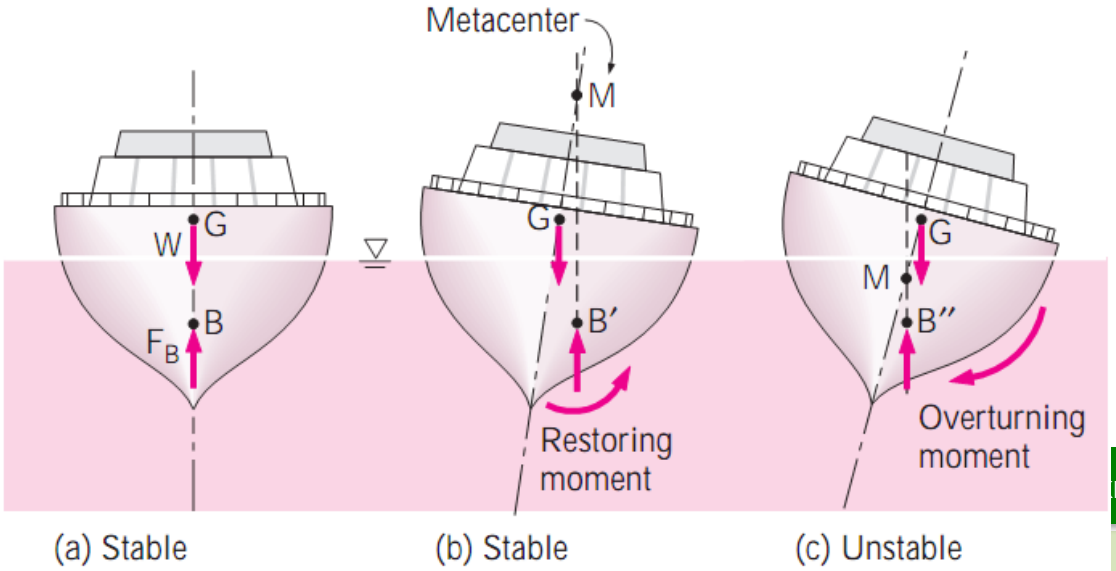
- If a floating body is bottom-heavy and thus the center of gravity G is directly below the center of buoyancy B, the body is **always stable**.

- But unlike immersed bodies, a **floating body** may **still be stable** when **G is directly above B**.



Stability of Floating Bodies

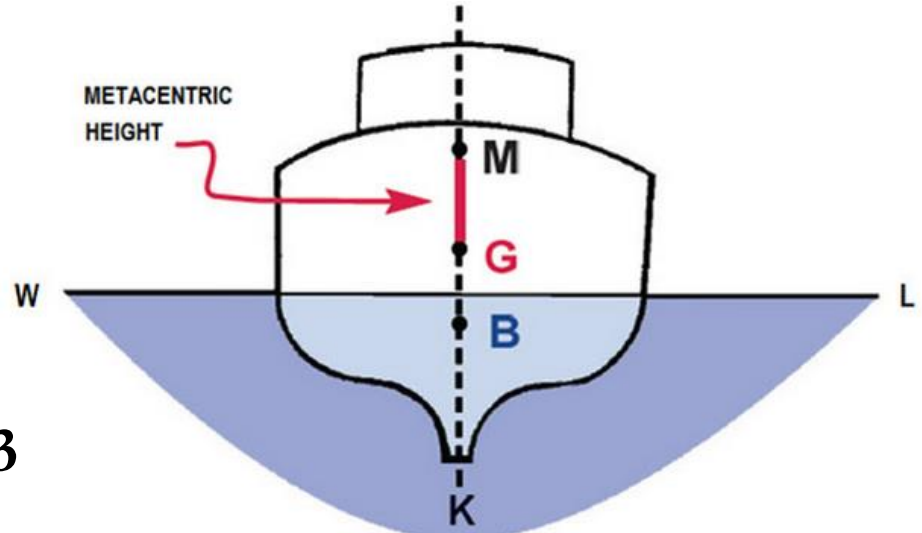
- The **centroid** of the displaced volume shifts to the side to a point **B'** during a rotational disturbance while the center of gravity G of the body remains unchanged.
-
- If point **B'** is sufficiently far, these two forces create a restoring moment and return the body to the original position'.

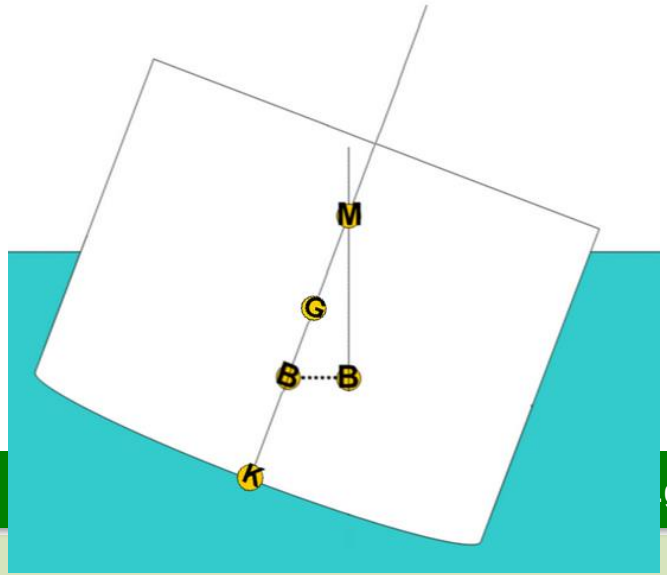


Stability of Floating Bodies

- The **metacentre** (M) is the point where the **line of action of the buoyant force** through the **centre of gravity** of the body intersects the line of action of the buoyant force **after rotation**.

- **Metacentric height, GM**, is the distance between the center of gravity G and the metacenter M.



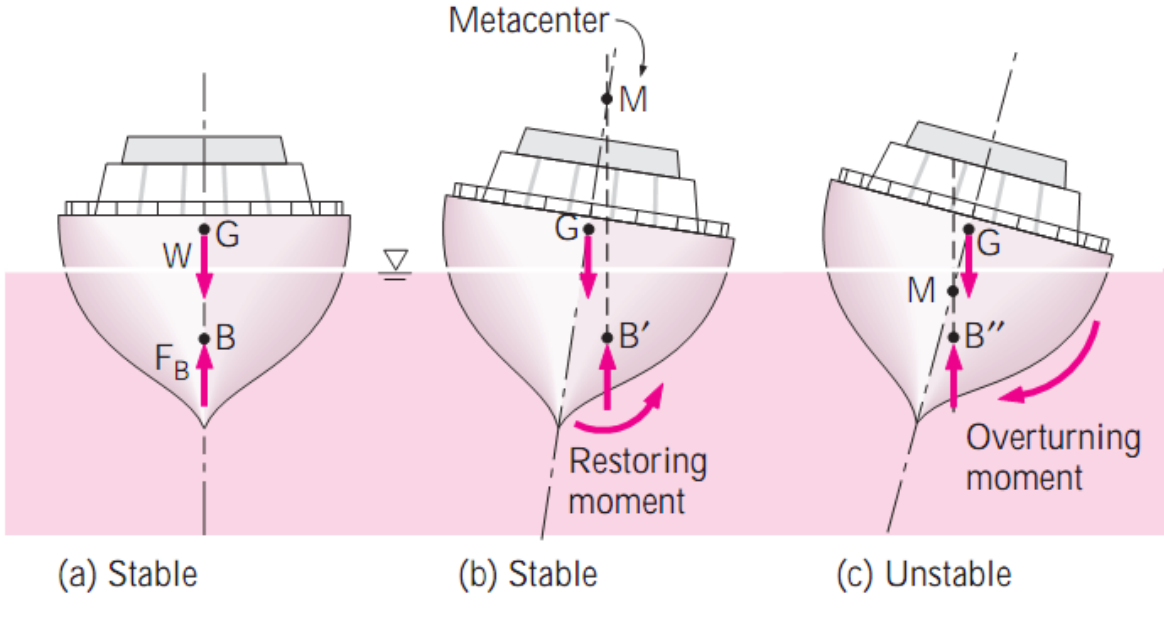


Stability of Floating Bodies

A floating body is:

- Stable if **point M** is above **point G**, (GM is positive)
- Unstable if point M is below point G, (GM is negative).
  - the weight and the buoyant force acting on the tilted body generate an overturning moment instead of a restoring moment, causing the body to capsize.

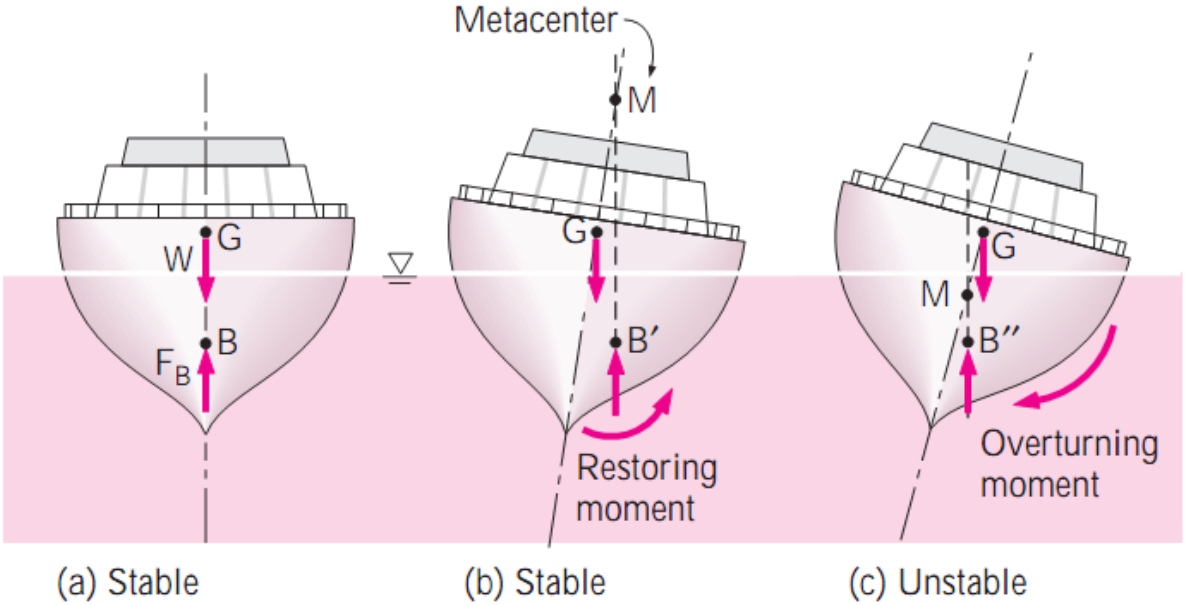




Stability of Floating Bodies

- The length of the metacentric height (GM) above G is a measure of the stability
- Larger GM is, the more stable is the floating body





Thank You!